# Assignment 1

Luke Milne

Student Number: 10566650

CSG1105

Instructor: Don Griffiths

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# Introduction:

This report is an analysis of network traffic using Wireshark. The aim of Task 1 is to investigate how web traffic is established and transmitted using HTTP over TCP. Task 2 examines the SMTP communication within a provided packet capture file. The objective of the assignment is to see if we can identify and analyse DNS, TCP and HTTP traffic, understand TCP three-way handshake, examine SMTP connection and investigate security improvements in email protocols. Wireshark was used to capture live traffic and analysing pre-recorded traces in the assignment.

# Methodology:

The tools used were Wireshark, Web browser(incognito mode), HTTP site( http://www.faqs.org/faqs ) and a SMTP trace file( smtp.pcapng).

Task 1:

This started with me opening a packet capture on wireshark, then I would go to the HTTP site http://www.faqs.org/faqs in incognito mode. Once the site has loaded I stop the capture and start to apply filters to my scan.

1. Type “dns” in the filter bar
2. Once filtering by dns requests allows me to find the IP address of the HTTP site and what DNS server was used to resolve the FQDN to an IP address.
3. I then filter for traffic on this IP address
4. You look for packets where the info has SYN, then SYN,ACK and ACK one after the other
5. You click on the 1st SYN packet and go into TCP and locate the Destination Port number
6. Filter the traffic to only getting HTTP  traffic from the destination IP address
7. Scroll down the scan and look for a packet whose info is GET /faqs.
8. Once in the GET packet expand the HTTP section then the GET /faqs section and you will find the request method there.
9. Carry on looking down the scan for a packet that says HTTP/1.1 200 OK. Open that up and the HTTP section and you find the status code 200
10. 1st go to the GET request packet and look inside the HTTP section that is where I find all the info. Then I click the response in frame link which will take me to the response packet, this will show me all the info for this header.
11. Refer to Results
12. I go to the bottom of the GET request packet, there is a hyperlink that takes you to the response packet.
13. Look in the response packet and open the HTTP section, the server header should be there.
14. Open up the response packet and open the HTTP section , the content-type is there.
15. Look through the packets for a HTTP GET request that has /image/ in the info section. Then go into HTTP and click on the response in frame. Open the HTTP of that packet and you will find the content-type.

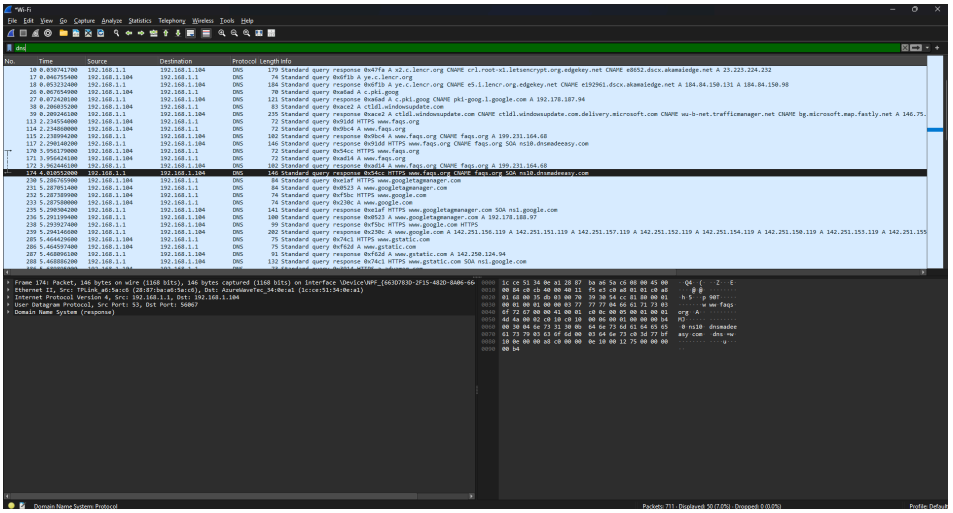
Task 2:

Process of connecting to the SMTP server and sending content:

- The server greeted me, the server was ready and the connection was established.
- The Client identifies itself, EHLO
- The server tells the client what is supports
- The server asks for the username and password, which gets sent through and confirmed if correct
- The sender and recipient are defined and the email is sent
- The email is confirmed to be received by the server
- The session ends

# Results:

## Task 1:

1. 

2. a.) The DNS query was resolved to the IPv4 address of 199.231.164.68 and the DNS server used to resolve the FQDN was 192.168.1.1.

| | | | | | | |
|---|---|---|---|---|---|---|
| 114 | 2.234860000 | 192.168.1.104 | 192.168.1.1 | DNS | 72 | Standard query 0x9bc4 A www.faqs.org |
| 115 | 2.238994200 | 192.168.1.1 | 192.168.1.104 | DNS | 102 | Standard query response 0x9bc4 A www.faqs.org CNAME faqs.org A 199.231.164.68 |

3. Ip.addr = 199.231.164.68 is the filter I used

4. The purpose of the 3 way handshake is to ensure reliable communication before you transfer data. SYN initiates connection, SYN-ACK is the server acknowledges and agrees then ACK the client confirms and establishes connection.

| | | | | | | |
|---|---|---|---|---|---|---|
| 173 | 3.969615100 | 192.168.1.104 | 199.231.164.68 | TCP | 66 | 51663 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM |
| 178 | 4.184476700 | 199.231.164.68 | 192.168.1.104 | TCP | 66 | 80 → 51663 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128 |
| 179 | 4.184570300 | 192.168.1.104 | 199.231.164.68 | TCP | 54 | 51663 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0 |

5. b.) The port number is 80. This will change every time you visit the site

```
Source Port: 63214
Destination Port: 80
[Stream index: 18]
```

6. http && Ip.addr = 199.231.164.68 is the filter used

7. This is the get package i chose

| | | | | | | |
|---|---|---|---|---|---|---|
| 185 | 4.618978900 | 192.168.1.104 | 199.231.164.68 | HTTP | 520 | GET /faqs/ HTTP/1.1 |

8. c.) The GET method is used in the captured traffic. This is my reference for all the HTTP methods. (Mozilla Developer Network, n.d)

9. d.) 1xx means informational, 103 the server returns some response headers before the final HTTP response is sent. 2xx means success, 201 The request was fulfilled and the server created a new resource. 3xx means Redirection, 301 is when the URL of the requested resource has been replaced with a new address. 4xx means Client errors, 404 The URL doesn't exist. 5xx means server error, 505 the server doesn't support the HTTP protocol version used in the request. (SiteGround, n.d.)

10. e.) The request headers include fields such as Host, Accept and Connection. The Response header includes fields like server, content-type and Date. You can find the definition of these headers at the IETF's website, it is RFC 723. (Internet Engineering Task Force [IETF], 2014)

11. f.) The client sends HTTP GET requests then the server processes that request, once the server has processed and is Okay it sends a HTTP response 200 and then the browser renders content.

12. 187 4.835583400 199.231.164.68 192.168.1.104 HTTP 1471 HTTP/1.1 200 OK (text/html)

13. g.) The website uses Apache, so other common servers are Nginx and Microsoft IIS

14. h.) The content-type is text/html used for a web page and charset = UTF-8 defines the character encoding used to display the text correctly on the website, UTF-16 is different because it typically uses 2 or more bytes per character, making it less efficient for english text. Without the Content-Type header, the browser would not know how to interpret the data correctly

15. The content-type is image/png, its purpose is for images, CSS and JS files. The MIME type is a standard that describes file types on the internet the types are text/html, image/png and application/json. The IANA defines all MIME types. (Mozilla Developer Network, n.d.)

<u>Task 2:</u>

1. SMTP is a push-based protocol designed specifically for sending emails between servers. HTTP is a request/response protocol used to retrieve web content. SMTP is a stateful, command-driven protocol where a session is maintained by commands such as HELO, MAIL FROM and DATA. HTTP is stateless and each request is independent, SMTP normally uses port 25 while HTTP normally uses port 80. (GeeksforGeeks, n.d.)

2. SMTP uses a client-server architecture where a sending mail server initiates connection to a receiving mail server. The client is responsible for establishing TCP connections and issuing commands. The server listens for incoming connections and responses with status codes. (GeeksforGeeks, n.d.)

3. SMTP, POP and IMAP work together to make email fully functional. SMTP is used to send emails from a client to mail server and between mail servers. POP downloads emails to a single device making it good for offline access. IMAP allows emails to be synchronised across multiple devices, enabling consistent access from different locations. (GeeksforGeeks, n.d.)
4. SMTP and IMAP transmit data in plain text, making them vulnerable to eavesdropping and interception. To enhance security you can add encryption using TLS and SMTP can now use STARTTLS. (Cloudflare, n.d.

# Conclusion and Reflection:

This assignment provided valuable insight into how network communication operates in real-world environments. Task 1 enhanced my understanding of how web traffic is established and maintained through protocols such as TCP and HTTP. While the process initially appears simple from a user perspective, analysing the packet-level data in Wireshark revealed the complexity involved, including multiple layers of communication, protocol interactions, and control mechanisms such as the TCP three-way handshake.

Task 2 helped develop my understanding of email communication using SMTP. I learned how emails are sent and received using SMTP and the commands used between the client and server. It showed me how important POP and IMAP are to having a fully functional email service and how important security is when transferring data.

Overall, this assignment increased my skills at analyse network traffic and interpret protocol behaviour. If I were to approach this task again, I would improve by documenting sources throughout the process and structuring the report prior to completing the video presentation. This would help ensure better organisation, clearer explanations, and more efficient completion of the assignment.

# References: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